TEACHER RESOURCES | 2026

TEACHER HANDBOOK

Running a science fair unit in your classroom.

For teachers delivering the FairGame curriculum, Grades 5 through 8

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START HERE

This handbook is the classroom companion to the FairGame Cross-Curricular Science Fair Curriculum. The curriculum tells you what happens in each week. This handbook tells you how to run it, how to grade it, and what to do when a student falls behind.

TWO DESIGN PRINCIPLES

The science fair is the project

Every assignment is a real deliverable students use in competition, not busy work added on top of existing coursework. Nothing in this unit is practice for something else.

The journal is the safeguard

Science journals and in-class checkpoints are the primary protection against fabricated work. Students who keep a genuine record of their process cannot invent it later. Journal collection every three weeks is non-negotiable at all grade levels.

WHAT YOU NEED BEFORE DAY ONE

- Your school's fair date, working backward from the OAS registration deadline
- A class set of science journals, one per student, that stay at school between check-ins
- The grade-band timeline for your students, in the Grade Module for your grade
- The four grading rubrics, one for each project stage
- Agreement from the other teachers on your cross-curricular team about who covers what

> **YOU DO NOT NEED A SCIENCE BACKGROUND**
>
> A science fair unit can be run by any motivated teacher. FairGame provides the content knowledge through these resources; you provide the organizational energy. Many successful fairs are coordinated by teachers outside the sciences.

THE CROSS-CURRICULAR MAP

The work spreads across classrooms because good science happens across disciplines. Teachers do not need to coordinate every lesson, but each teacher should know what the others are working on so students hear consistent vocabulary and a consistent timeline.

PHASE	SCIENCE	ELA	MATH	SOCIAL STUDIES	ART / ELECTIVE
Asking the Question	Testable question, variables, OAS categories	Reading science texts, evaluating sources	Statistical vs non-statistical question	Context of the problem	Sketchbook: initial brainstorm visuals
Background Research	Scientific method, credibility of evidence	Note-taking, summarizing, citing sources	Ratios and percentages in data contexts	Real-world application of the topic	Thumbnail sketches of display layouts
Experimental Design	Variables, controls, safety, procedure writing	Technical writing: step-by-step procedures	Sampling, number of trials, measurement tools	Historical or societal relevance	Mood board for display board aesthetic
Data Collection	Running the experiment, lab notebook entries	Daily journal entries, observation writing	Data tables, mean, range, graphing	Class focus is execution	Photography or sketching in progress
Analysis and Conclusion	Interpreting results, claim, evidence, reasoning	Conclusion paragraph writing and revision	Graph construction, trend description	Connecting results to real-world impact	Final board layout and typography
Presentation	Practice question and answer, teacher as judge	Oral presentation structure, public speaking	Explaining data verbally with precision	Audience awareness: who cares about this	Final display board design and assembly

IF YOU ARE THE ONLY TEACHER ON THIS

The map still works. Deliver the Science column in class, assign the ELA column as writing homework, and borrow the Math column during your data week. The other columns are enhancements, not requirements.

THE SCIENCE JOURNAL

Every student maintains a science journal from the first day of the unit through the final presentation. The journal is the paper trail teachers and judges rely on, and it travels with the student across classrooms. Any teacher on the team may write comments in it. At competition, students may show it to judges as evidence of authentic process work.

WHAT GOES IN THE JOURNAL

- Date and time of every work session, even short ones
- Written observations during every trial: what was seen, heard, or measured
- Questions that came up during the process, and how they were answered
- Sketches, diagrams, and draft data tables
- Notes from any interview, mentor conversation, or background reading
- A running log of hours worked on the project
- Any time the original plan changed, and why

COLLECTION SCHEDULE

COLLECTION	APPROXIMATE DATE	WHAT YOU REVIEW
Check 1	End of Week 3	Question is testable, hypothesis is written, at least one background source cited
Check 2	End of Week 6	Experimental design is complete, procedure is written, materials are identified
Check 3	End of Week 9	Data collection is underway or complete, data table has real entries
Check 4, final	End of Week 12	Complete record from start to finish, used as part of presentation evaluation

GRADE THE JOURNAL, NOT ONLY THE OUTCOME

Consider grading journal completeness at each check-in rather than grading final outputs alone. A student who kept a thorough journal and had a hypothesis the data did not support has done excellent science. A student with a perfect conclusion and an empty journal has not.

CHOOSING YOUR GRADE BAND

The curriculum runs at three levels of independence. Use the band that matches your students, not strictly their grade number.

GRADES 5 TO 6 — GUIDED EXPERIMENTS, FULLY IN CLASS

Students choose from a menu of pre-approved experiments completed entirely at school with teacher-provided materials. This removes the barrier of parental involvement and ensures every student can participate regardless of what is available at home. The first nine weeks walk all students through the process together. In the second nine weeks, teachers identify students ready for independent OAS entry.

GRADE 7 — OPEN TOPIC, MOSTLY IN CLASS

Students choose their own topic, which drives follow-through on a longer process. Structure keeps most work in class where teachers can monitor progress. Home experimentation happens only for projects needing time-based observation. Teachers move from directing to coaching.

GRADE 8 — STUDENT-LED, MILESTONE DRIVEN

Class time acts as a checkpoint rather than a teaching sequence. Students arrive having completed a deliverable, receive feedback, and leave with a next step. Students who fall behind the milestone schedule receive an immediate check-in conversation, not a grade penalty on the final project.

TIMELINE FLEXIBILITY

Ohio fairs typically run December through February for OAS regional and district events. Data collection must be complete before winter break. Build your check dates backward from your school's registration deadline, not forward from the first day of school. Schedule one buffer week before each major milestone; losing a week to testing or closures is expected and fine.

ASSESSMENT AND TROUBLESHOOTING

FOUR RUBRICS, FOUR STAGES

Grade at four points rather than once at the end. Each rubric is a separate document in this folder and uses the same four-level scale, so scores are comparable across stages.

STAGE	RUBRIC	WHEN
1	Project Proposal Rubric: question, hypothesis, design	Weeks 2 to 6 by grade band
2	Progress Check Rubric: journal, data collection, adaptation	Weeks 6 to 9 by grade band
3	Final Project Rubric: results, conclusion, report, board	Weeks 11 to 13 by grade band
4	Presentation Rubric: delivery, understanding, questions	Weeks 12 to 14 by grade band

COMMON PROBLEMS AND WHAT TO DO

A student's hypothesis is flawed after design review

Let them revise it before data collection begins. This is not a failure, it is how science works. Build a formal hypothesis check into design approval.

A procedure is producing unreliable results mid-collection

Students may adjust the design during data collection. Changes must be documented in the journal with a dated explanation.

A student says their project is not scientific enough without lab equipment

Everyday materials produce valid data for most middle school projects: rubber bands, cardboard, measuring cups, pharmacy thermometers, kitchen scales. Say this out loud early and often.

A student falls behind the milestone schedule

Hold an immediate check-in conversation. Do not apply a grade penalty to the final project for a missed interim date.

You suspect AI-generated writing

The journal is your baseline. Compare the report against dated journal entries and short in-class writing done during check-ins.

A student picks a topic outside your expertise

Say so honestly and figure it out together. Students who choose topics outside the teacher's expertise are doing harder science.

THE OAS PATHWAY

The Ohio Academy of Science runs the primary competitive pathway for Ohio students. Regional and district fairs run December through February, with State Science Day held annually in Columbus. Every deliverable in this curriculum maps directly to an OAS submission requirement.

OAS REQUIRES	CURRICULUM DELIVERABLE	READY BY
Testable research question	Milestone 1, or Lesson 2 in Grades 5 to 7	Weeks 1 to 2
Written hypothesis, if-then format	Hypothesis Worksheet, Milestone 3	Weeks 3 to 6
Replicable experimental procedure	Experimental Design Planner, Milestone 3	Weeks 4 to 6
Data table, multiple trials and averages	Journal entries, Data Collection Sheets	Weeks 7 to 10
Graphs of results	Math class graph construction, Milestone 7	Weeks 9 to 11
Written conclusion in CER format	Conclusion writing lesson, Milestone 8	Weeks 10 to 12
Research paper	Research Paper Template, Milestone 8	Weeks 11 to 13
Display board, 48 by 36 inches	Display Board Template, Milestone 9	Weeks 12 to 13
Oral presentation to judges	Practice Fair, Milestone 10	Weeks 13 to 14

OAS registration happens through ProjectBoard at oas.projectboard.net. Allow 45 minutes to walk through registration for your school the first time.

FREE SUPPORT FROM FAIRGAME

Partner schools receive the full teacher resource library, OAS registration support, judge recruitment assistance, and mentor matching through the Industry Mentorship Program. Everything is free.

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